

Disrupción, Innovación y Tecnología

Clase 4
17 de marzo

Dr. Juan M. del Nido
Primer semestre 2025

Recapitulando

Magia, carisma, promesas, sublime, el
cambio, nuevo futuro...

Quién?

Hoy: Los Disruptores



Los Disruptores

La emergencia de la figura del disruptor en nuestra cultura

sus virtudes,

su ética,

su relación con el entorno,

su relación con el sublime, la magia, etc.

Royal Society of London for Improving Natural Knowledge



Joseph Banks (1743-1820)

Hijo mayor de una familia rica

Educado en Eton y Oxford

Naturalista

Compañero de viajes de James Cook

“Endeavour: Una iniciativa conjunta entre el Almirantazgo y la Royal Society para observar el tránsito de Venus de 1769 desde Tahití. Una observación exitosa constituiría un triunfo nacional y aportaría conocimientos útiles para la navegación. El consejo de la Royal Society solicitó que Joseph Banks, miembro de la Sociedad, caballero y experto en historia natural, también se embarcara en el viaje.”

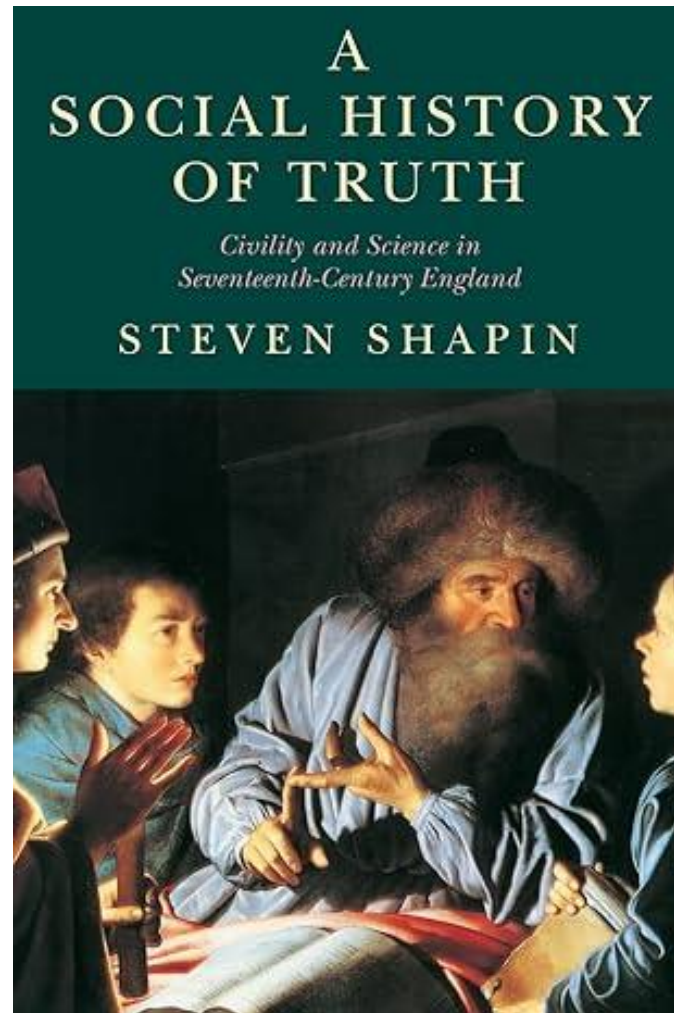
“La Royal Society (...) era una sociedad de caballeros, por razones tanto filosóficas como pragmáticas. Filosóficamente, su condición de caballeros garantizaba el conocimiento que producían (...).

Pragmáticamente, la sociedad dependía de que sus líderes sepan moverse en la elite de la sociedad inglesa. Para la década de 1760, la Royal Society era parte integral de una red de patrocinio e influencia mutuos que incluía departamentos de estado como el Almirantazgo, instituciones culturales como el Museo Británico y empresas comerciales como la Compañía de las Indias Orientales.”

resultado: conocimiento práctico, caballeresco

El establishment: una ciencia exclusiva

“Los problemas de credibilidad en la ciencia se resolvieron prácticamente a través de los códigos y convenciones de conducta gentil: confianza, civilidad, honor e integridad.”

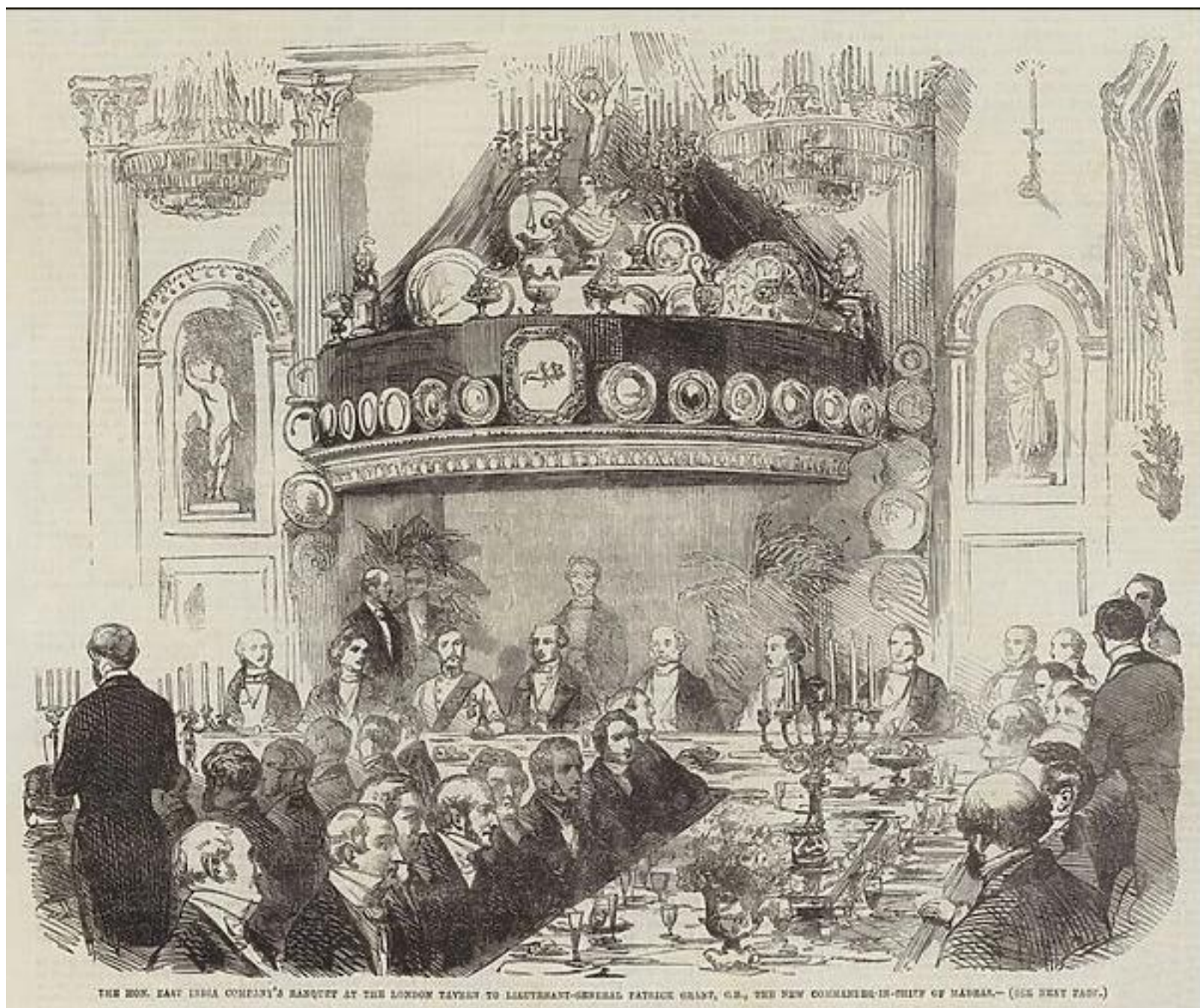


Naturalismo, filosofía natural, filosofía experimental...

conocimiento principalmente como adorno para caballeros cultos (...) que también podría ofrecerles un beneficio económico sustancial a través de planes para la mejora agrícola o la explotación colonial

Big picture, sin chiquitaje

Ejemplo



THE HON. EAST INDIA COMPANY'S BANQUET AT THE LONDON TAVERN TO LIAUTENANT-GENERAL PATRICK GRANT, G.B., THE NEW COMMANDER-IN-CHIEF OF MADRAS.—(SEE NEXT PAGE.)

1820 muere Banks

La Taberna de los francomasones



Los Geólogos y los Astrónomos

Una ciencia nueva y más disciplinada. Coincidían en que la ciencia era cosa de caballeros, pero tenían opiniones muy diferentes sobre qué debían hacer los caballeros para convertirse en hombres de ciencia. **La clave era la autodisciplina.** Creían que sus enemigos eran amateurs corruptos y que la membresía de la Royal Society debía otorgarse por lo que sabían, no por a quién conocían..

La utilidad de la ciencia residía en los detalles finos
– el chiquitaje. Recopilaban sistemáticamente
conocimientos geológicos y astronómicos, lo cual
requería minuciosidad y esfuerzo, y no era para
quienes tenían un interés superficial en la ciencia.
Solo podía lograrse mediante el esfuerzo conjunto
de quienes realmente trabajaban los temas.

Un nuevo tipo de conocimiento: no caballeroso,
conocimiento especializado y disciplinado.

Conocimiento: algo útil en sí mismo, pero también útil
por su capacidad para generar progreso social y
económico.

Se corre la filosofía de lugar

Primes.

0

32.

285	6451	3.80962	70418	94049	73044	71154	009
285	6469	3.81083	71511	40488	32105	18595	291
443	6473	3.81110	56070	17938	39638	82884	750
443	6481	3.81164	20214	53151	00957	52230	112
443	6491	3.81231	16091	31123	77347	13412	138
444	6521	3.81431	42002	07459	56939	61526	137
444	6529	3.81484	66686	04463	28879	38208	654
445	6547	3.81604	23409	21996	61877	22757	457
290	6551	3.81630	75994	31939	80046	72301	026
445	6553	3.81644	01679	56138	66098	46667	768
445	6563	3.81710	24042	56923	14873	38882	086
446	6569	3.81749	92618	67758	27469	60352	877
446	6571	3.81763	14671	90515	35672	81171	829
447	6577	3.81802	78418	59256	21372	18985	921
356	6581	3.81829	18907	99995	91567	93449	242
447	6599	3.81947	81283	62122	59953	90283	597
448	6607	3.82000	43068	08317	90132	58509	960
281	6619	3.82079	23810	88203	71812	51443	937
448	6637	3.82197	18176	42042	81425	92801	810
448	6653	3.82301	75234	46049	24015	85578	466
250	6659	3.82340	90148	92544	58225	74521	388
449	6661	3.82353	94336	56858	99207	36604	562
449	6673	3.82432	11248	50771	26538	39903	406
449	6679	3.82471	14434	64734	31807	23109	428
411	6689	3.82536	11959	52633	33507	08708	128
450	6691	3.82549	10298	79430	87742	40028	456
450	6701	3.82613	96179	35914	76357	41853	291

TABLE OF SIMPLE SUBSTANCES.

Simple substances belonging to all the kingdoms of nature, which may be considered as the elements of bodies.

	<i>New Names.</i>			<i>Correspondent old Names.</i>
Light	-	-	-	Light.
Caloric	-	-	-	Heat.
				Principle or element of heat.
				Fire. Igneous fluid.
				Matter of fire and of heat.
Oxygen	-	-	-	Dephlogisticated air.
				Empyrean air.
				Vital air, or Base of vital air.
Azote	-	-	-	Phlogisticated air or gas.
				Mephitic, or its base.
Hydrogen	-	-	-	Inflammable air or gas,
				or the base of inflammable air.

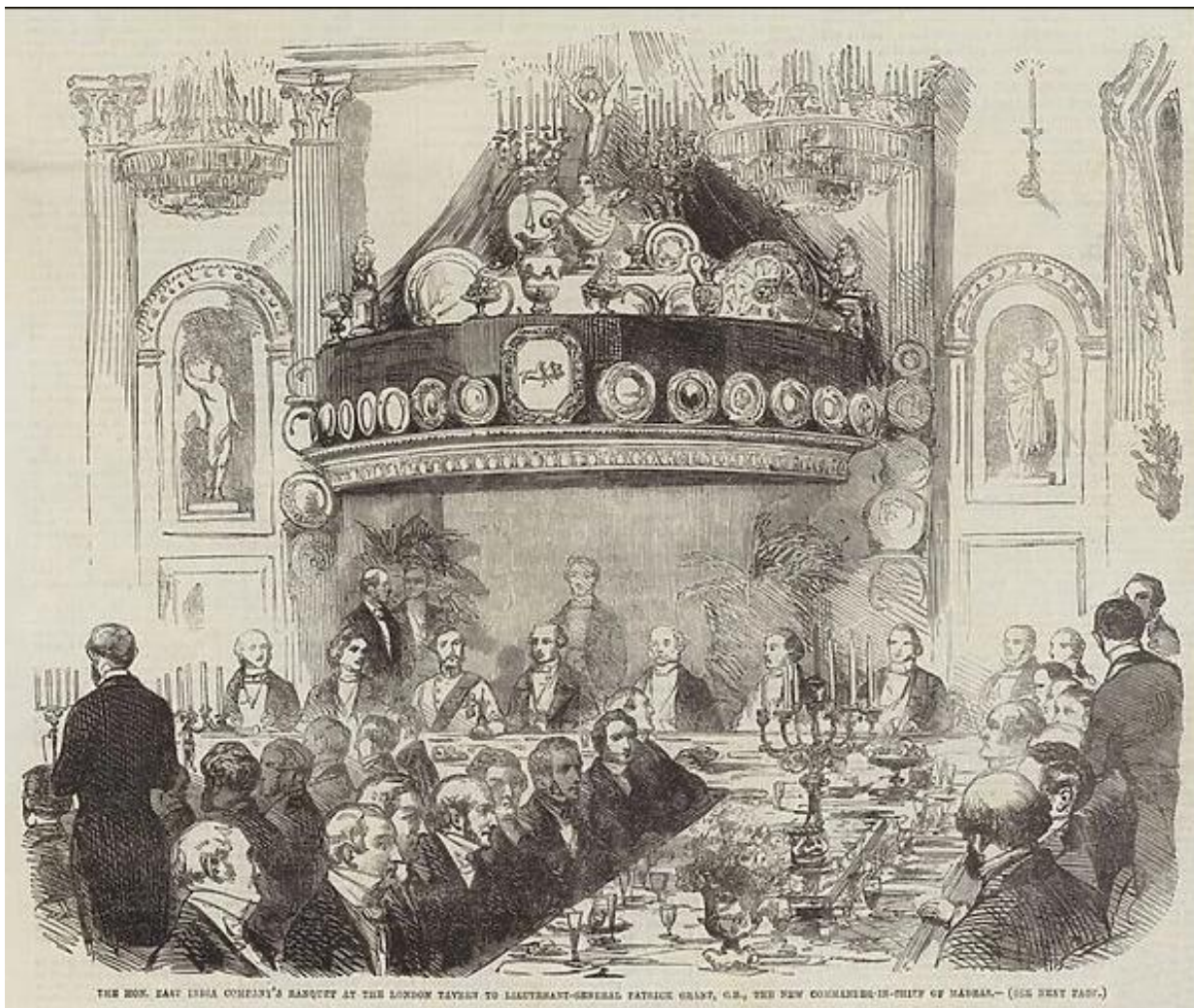
Oxydable and Acidifiable simple Substances not Metallic.

	<i>New Names.</i>			<i>Correspondent old names.</i>
Sulphur	-	-	-	The same names.
Phosphorus	-	-	-	
Charcoal	-	-	-	
Muriatic radical	-	-	-	Still unknown.
Fluoric radical	-	-	-	
Boracic radical	-	-	-	

Oxydable and Acidifiable simple Metallic Bodies.

	<i>New Names.</i>			<i>Correspondent Old Names.</i>
Antimony	-	-	-	Antimony.
Arsenic	-	-	-	Arsenic.
Bismuth	-	-	-	Bismuth.
Cobalt	-	-	-	Cobalt.
Copper	-	-	-	Copper.
Gold	-	-	-	Gold.
Iron	-	-	-	Iron.
Lead	-	-	-	Lead.
Manganese	-	-	-	Manganese.
Mercury	-	-	-	Mercury.
Molybdena	-	-	-	Molybdena.

Regulus of



THE HON. EAST INDIA COMPANY'S BANQUET AT THE LONDON TAVERN TO MAJOR-GENERAL PATRICK GRANT, G.B., THE NEW COMMANDER-IN-CHIEF OF HAVANA.—(SEE NEXT PAGE.)

Primes. 0 32.

285	6451	3.80962	70418	94049	73044	71154	009
285	6469	3.81083	71511	40488	32105	18595	291
443	6473	3.81110	56070	17938	39638	82884	750
443	6481	3.81164	20214	53151	00957	52230	112
443	6491	3.81231	16091	31123	77347	13412	138
444	6521	3.81431	42002	07459	56939	61526	137
444	6529	3.81484	66686	04463	28879	38208	654
445	6547	3.81604	23409	21996	61877	22757	457
290	6551	3.81630	75994	31939	80046	72301	026
445	6553	3.81644	01679	56138	66098	46667	768
445	6563	3.81710	24042	56923	14873	38882	066
446	6569	3.81749	92618	67758	27469	60352	877
446	6571	3.81763	14671	90515	35672	81171	829
447	6577	3.81802	78418	59256	21372	18985	921
356	6581	3.81829	18907	99995	91567	93449	222
447	6599	3.81947	81283	62122	59953	90283	597
448	6607	3.82000	43068	08317	90132	58509	960
281	6619	3.82079	23810	88203	71812	51443	937
448	6637	3.82197	18176	42042	81425	92801	810
449	6653	3.82301	75234	46049	24015	85578	466
250	6659	3.82340	90148	92544	58225	74521	388
449	6661	3.82353	94336	56858	99207	36604	562
449	6673	3.82432	11248	50771	26538	39903	406
449	6679	3.82471	14434	64734	31807	23109	428
411	6689	3.82536	11959	52633	33507	08708	128
450	6691	3.82549	10298	79430	87742	40028	456
450	6701	3.82613	96179	35914	76357	41853	291

Recreo

1827

Los hombres prácticos: Marc e Isambard Brunel y el Túnel del Támesis





THE COMPANY'S TREASURER ON HER JOINTURE BARREL, SURROUNDED BY ONE OF THE TUNNEL RELATIONS.

When the ingenious Mr. Fawcett visited Constantinople in 1826, he was not long surprised that project so being abandoned by an Albanian chief as to the Thames Tunnel. This, however, is but one of the many instances of the anxiety with which the great work has been regarded throughout Christian Europe. In Egypt, where a new country is rising, pharaohs like, even the idea of the old world, the project of the Tunnel has been regarded with unusual interest—perhaps, too, through the excited state. But not without inquiry extended: "the project of the world" in every time since, the Thames Tunnel would long have been the subject of inquiry, for the first attempt to effect a communication between the shores of a wide and deep river, without any interruption in its navigation, has had, and probably will have, an powerful for many ages. When and where was likely again to combine the mechanical, physical and economical; the resources, presented and scientific; the natural laws of communication; and the courage and energy necessary to the successful prosecution of such a work? And also answer, "What?" Inevitably as here, the difficulties, which it has required almost superhuman courage to overcome, the great work has been achieved; and an admirer has the communication was witnessed by standing themselves.

A tunnel (from Constantinople to Tilbury) under, in the chief instance, in the present work, was proposed and even commenced by Mr. Rennie in 1790, but was abandoned as impracticable. Two or three years after this, an attempt was made to make the present Tunnel, to connect Southampton and London, by an experiment (which failed, indeed) of a shaft was sunk 50 feet, and the horizontal connection made in the form of a straggling line, afterwards widened into a passage, and carried to within 100 feet of the Middlesex canal, when the engine gave up the work as impracticable, notwithstanding certain were offered for them, and many more received. It was under these circumstances, however, that, in 1825, Mr. James W. & J. Rennie, engaged a large sum, which involved the sanction of many gentlemen of rank and station, among whom were the Duke of Devonshire and the Marquis of Salisbury. The project between Southampton and Weymouth, which the proposed communication, in perhaps, the only case where London Bridge and Greenwich where such a roadway could have been constructed without interfering seriously with some of the great commercial establishments on both sides of the river.

The importance of this new mode of communication between

two such places, only now 1200 feet apart respectively (but four miles by the way of London Bridge), will be at once apparent. But it is still more so, if we consider for a moment the position



MR. A. RENNIE.

24th August 1825 during his progress through the Tunnel.

between the two great interests which belong to the different sides of the river. An immense amount of the foreign goods brought into the West India, the Levant, and St. Katherine's Docks, on the north side, is absorbed by the coasting-trade on the south side, it appears, in almost entirely converted form into the other by land-carriage? (Hagley's "London.") The communication of the Tunnel to passengers is shown by 5,700 persons transported daily by the different boats in the neighbourhood. There will be an extensive approach to the Tunnel to some; on the north, it is connected through Old London-bridge, with Southwark-bridge, and a new road is projected in continuation of the former, to the Commercial-road, Whitechapel; on the south it is close to the Deptford Lower-road.

Early in 1826 a company was formed, and in June an act of Parliament was obtained, for executing the Tunnel, Mr. Rennie being appointed engineer. His works commenced previous, notwithstanding the opposition, had taken three months. Having located the bed of the Thames, in the direction of the proposed tunnel, and then very honourable report induced Mr. Rennie to go to work in a successful, better way than he had otherwise intended. He began, in March, 1826, by preparing for a shaft 50 feet in diameter, which he commenced at 100 feet from the Middlesex bank of the river. This he effected by constructing, on the northern of the ground, a brickwork cylinder of the above diameter, 40 feet in height and 1 foot thick. Great care he set up the steam-engine for pumping out the water, and for raising the earth to be taken from within the cylinder, which he then sunk, as shown, into the ground, in the way that will be described in the next part. By this means he passed through a bed of gravel and sand, 30 feet deep, full of broken stones, and, in fact, a quagmire, in which deep-water formerly had been employed to support their work. When the shaft was sunk to the present depth of 40 feet, another shaft, of 30 feet diameter, was sunk still lower, 10, at the depth of 50 feet, the ground suddenly gave way, sinking around the shaft, and water was thrown up with great violence. The existence of this quagmire had been foreseen by some eminent geologists, and the experience proved of essential service to the engineer.

The shaft and entrance having been completed, the horizontal excavation, at the Tunnel itself, was commenced at the depth of 10 feet, 10 feet in breadth, and 10 feet in height; the excavation being larger than the interior of the old Roman of Caenotum. It was to be defended by strong walls, and to have passageways for a



SCHEMATIC OF THE TUNNEL.

Argumentaban que podían lograrlo gracias a su experiencia práctica con máquinas, su tozudez y su dedicación. Los periódicos los convirtieron en héroes. Los hombres prácticos que se forjaron a sí mismos y los caballeros de la ciencia (...) se encontraban cada vez más en el mismo tipo de lugares.

El hijo de Isambard Brunel comentó sobre la relación de su padre con su abuelo: «Le debía mucho, no solo **la herencia de numerosos dones naturales** y una **formación profesional**, sino también un ejemplo de **previsión** y **perseverancia**, la única forma de asegurar el éxito de los grandes proyectos». Ese tipo de historia de **autodisciplina** y **confianza en uno mismo** fue fundamental para la percepción que estos hombres tenían de sí mismos. También fue fundamental para la percepción que los demás tenían de ellos. Los hombres prácticos fueron los héroes de la época victoriana.

Mechanic's Magazine, Museum, Register, Journal, & Gazette.

— Industry! rough power!
Whom labour still attends, and sweat, and pain ;
Yet, the kind source of every gentle art,
And all the soft civility of life.
Thomson.

No. 1.]

SATURDAY, AUGUST 30, 1823.

[Price 3d.]

JAMES WATT.



MEMOIR OF JAMES WATT,
THE GREAT IMPROVER OF THE

instruct the present and all future
generations of mechanics; and,
therefore, his biography has been

Trabajo duro y autodisciplina

James Watt fue un gran ingeniero (...) no sólo por ser un gran pensador, sino por su dedicación constante. Muchos «hombres de su época sabían más que Watt, pero ninguno se esforzó tanto como él por aplicar todo lo que sabía **a fines prácticos**»

Estos hombres prácticos se hicieron a sí mismos, sin educación formal. Aunque su idea de autodisciplina es distinta a la de los hombres de ciencia, compartieron una cultura de exactitud y precisión: la ciencia y la ingeniería se volvieron separadas pero codependientes. Ambos buscaban hacer valer su experiencia en el poder político para poder participar activamente en la construcción del futuro.

Cómo era el carácter, la idiosincracia del hombre práctico?

1859

SELF-HELP;
WITH ILLUSTRATIONS OF
CHARACTER AND CONDUCT.

By SAMUEL SMILES,
AUTHOR OF 'THE LIFE OF GEORGE STEPHENSON.'

"This above all,—To thine own self be true;
And it must follow, as the night the day,
Thou canst not then be false to any man."
SHAKESPEARE.

NEW EDITION, REVISED AND ENLARGED.

TWENTIETH THOUSAND.

LONDON:
JOHN MURRAY, ALBEMARLE STREET.
1860.

The right of Translation is reserved.

(Smiles) se hizo una reputación explicando a jóvenes las mejores maneras de superarse. Se inspiró en la religión evangélica, así como en la tradición de la independencia masculina, para promover la idea de que la autodisciplina y la perseverancia eran las claves del éxito, y los ingenieros eran los mejores ejemplos. (...) Las biografías de grandes ingenieros eran «casi equivalentes a los Evangelios: enseñaban una vida noble, un pensamiento elevado y una acción enérgica por su propio bien y el del mundo».

LIVERPOOL AND MANCHESTER RAILWAY. 1829

*The 'Rocket' built by Robert Stephenson
and winner of the Rainhill trials.*



gettyimages

Credit: UniversallImagesGroup

“Robert Stephenson había empezado como maquinista y luego como guardafrenos en una mina de carbón; aprendió a leer a los dieciocho años. A finales de sus veinte, se convirtió en mecánico de máquinas en la minas de carbón y aprendió por su cuenta los secretos de las máquinas de vapor. A los 24 años construyó su primera locomotora de vapor, y su primer ferrocarril a los 30. Encarnaba todo lo que un ingeniero debe ser, y todo su éxito fue fruto de su propio trabajo y su tenaz fuerza de carácter.”

iconoclastas, obsesivos y resueltos, autodidactas, autosuficientes; desconfiados de las elites intelectuales, sus invenciones se basaban en una determinación tenaz para hacer que las cosas funcionaran. Las virtudes de Edison eran las del pionero, solo que la frontera hacia la que se dirigía era la del futuro, de lo posible, que querían expandir haciéndola suya.

With its focus on individual virtue, the Victorian vision of the future is an exclusive one. When we subscribe to this paradigm about how — and by whom — the future is made, we're also relinquishing control over that future

1859

SELF-HELP;
WITH ILLUSTRATIONS OF
CHARACTER AND CONDUCT.

By SAMUEL SMILES,
AUTHOR OF 'THE LIFE OF GEORGE STEPHENSON.'

"This above all,—To thine own self be true;
And it must follow, as the night the day,
Thou canst not then be false to any man."
SHAKESPEARE.

NEW EDITION, REVISED AND ENLARGED.

TWENTIETH THOUSAND.

LONDON:
JOHN MURRAY, ALBEMARLE STREET.
1860.

The right of Translation is reserved.

1859

ON
THE ORIGIN OF SPECIES

BY MEANS OF NATURAL SELECTION,

OR THE
PRESERVATION OF FAVOURED RACES IN THE STRUGGLE
FOR LIFE.

By CHARLES DARWIN, M.A.,
FELLOW OF THE ROYAL, GEOLOGICAL, LINNEAN, ETC., SOCIETIES;
AUTHOR OF 'JOURNAL OF RESEARCHES DURING H. M. S. BEAGLE'S VOYAGE
ROUND THE WORLD.'

LONDON:
JOHN MURRAY, ALBEMARLE STREET.
1859.

The right of Translation is reserved.

SELF-HELP;

WITH ILLUSTRATIONS OF

CHARACTER AND CONDUCT.

By SAMUEL SMILES,

AUTHOR OF 'THE LIFE OF GEORGE STEPHENSON.'

"This above all,—To thine own self be true;
And it must follow, as the night the day,
Thou canst not then be false to any man."

SHAKESPEARE.

NEW EDITION, REVISED AND ENLARGED.

TWENTIETH THOUSAND.

LONDON:

JOHN MURRAY, ALBEMARLE STREET.

1860.

The right of Translation is reserved.

1859

ON

THE ORIGIN OF SPECIES

BY MEANS OF NATURAL SELECTION,

OR THE

PRESERVATION OF FAVOURED RACES IN THE STRUGGLE
FOR LIFE.

By CHARLES DARWIN, M.A.,

FELLOW OF THE ROYAL, GEOLOGICAL, LINNÆAN, ETC., SOCIETIES;
AUTHOR OF 'JOURNAL OF RESEARCHES DURING H. M. S. BEAGLE'S VOYAGE
ROUND THE WORLD.'

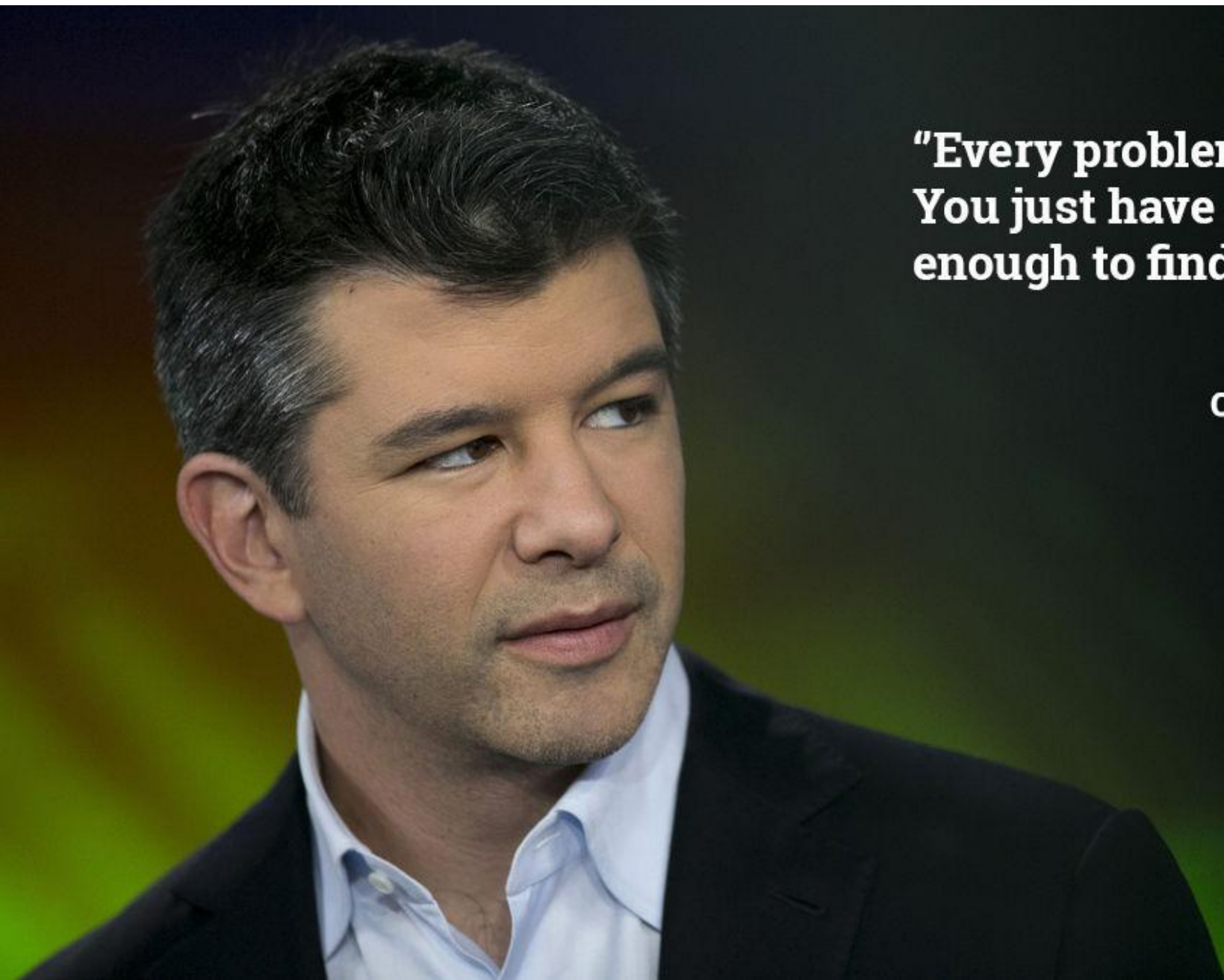
LONDON:

JOHN MURRAY, ALBEMARLE STREET.

1859.

The right of Translation is reserved.

Continuará...

A portrait of Travis Kalanick, co-founder of Uber, looking slightly to the right. He is wearing a dark suit jacket over a light blue shirt. The background is dark with a greenish-yellow glow.

**"Every problem has a solution.
You just have to be creative
enough to find it"**

**- Travis Kalanick,
co-founder of Uber**



**I believe you have to be willing to be
misunderstood if you're going to innovate.**

Jeff Bezos

*“Brilliant thinking is rare,
but courage is in even shorter
supply than genius.”*

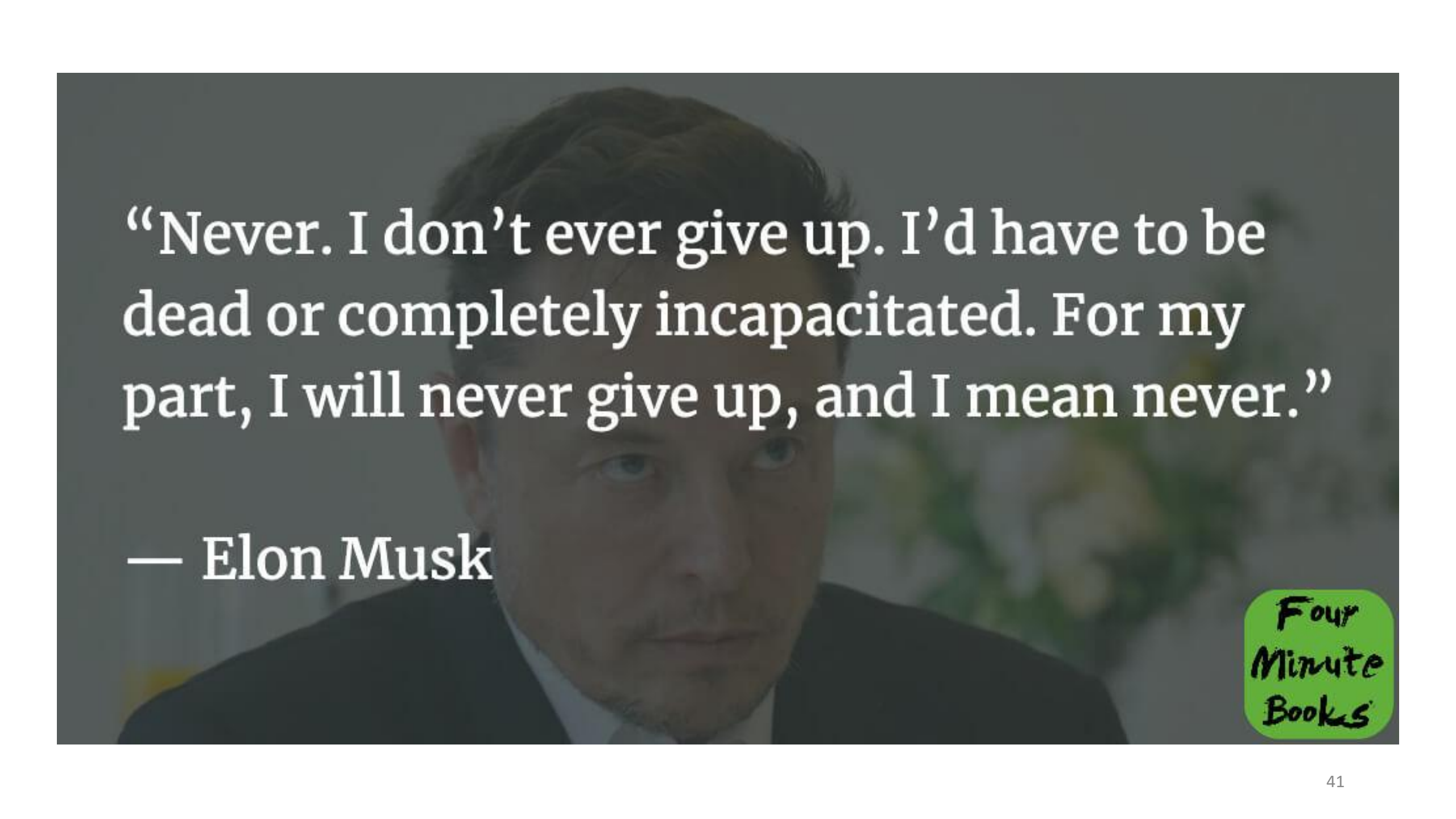
PETER THIEL



“We had a quote over the door that read,
**Success is not the result of
spontaneous combustion.
You must set yourself
on fire.”**

Elizabeth Holmes
31 year old billionaire
founder of Theranos

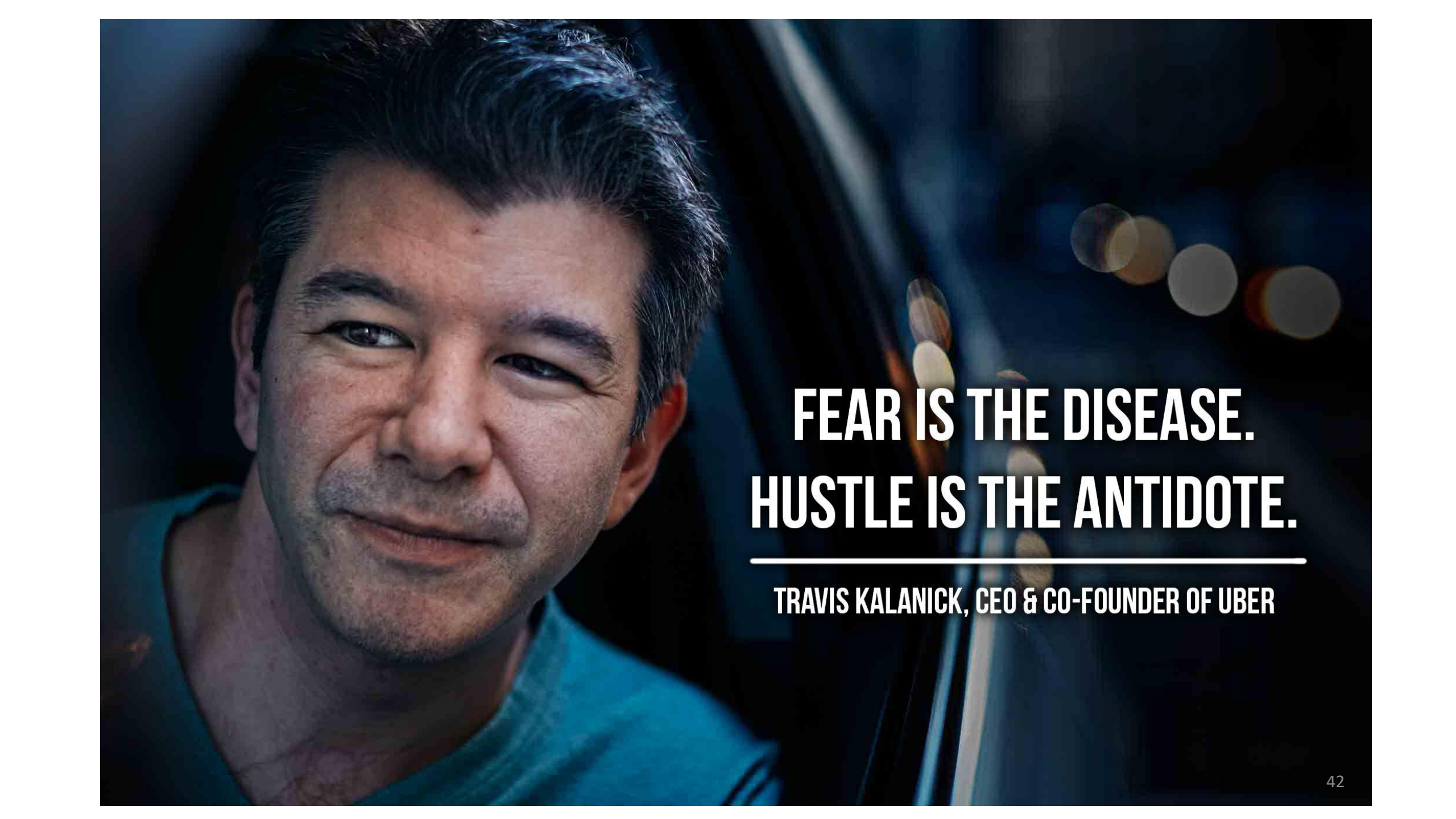




“Never. I don’t ever give up. I’d have to be dead or completely incapacitated. For my part, I will never give up, and I mean never.”

— Elon Musk

Four
Minute
Books

A portrait of Travis Kalanick, CEO and co-founder of Uber, is shown on the left side of the image. He is a middle-aged man with dark hair, wearing a teal shirt, and has a slight smile. The background is dark and out of focus, featuring some bokeh light effects. On the right side, there is a quote in white, bold, sans-serif capital letters. The quote is: "FEAR IS THE DISEASE. HUSTLE IS THE ANTIDOTE." Below the quote is a thin white horizontal line, and then the attribution "TRAVIS KALANICK, CEO & CO-FOUNDER OF UBER" in a smaller, white, sans-serif capital font.

**FEAR IS THE DISEASE.
HUSTLE IS THE ANTIDOTE.**

TRAVIS KALANICK, CEO & CO-FOUNDER OF UBER